
SPARSITY AND SUPERPOSITION IN MIXTURE OF EXPERTS

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ABSTRACT

Mixture-of-Experts (MoE) architectures have become the dominant paradigm for scaling large language models, yet how their internal representations diverge from those of the dense networks they increasingly displace remains poorly characterized. A mature line of work accounts for dense networks through *superposition*—the representation of more features than there are dimensions—and establishes that the extent of superposition is governed by feature sparsity and feature importance. Working in the controlled sparse-feature reconstruction setting in which this account was originally developed, we show that it does not carry over to MoEs: within this setting, neither feature sparsity nor feature importance induces the discontinuous phase changes characteristic of dense models, and it is instead *network sparsity*—the ratio of active to total experts—that governs the representational strategy an MoE adopts. To render this dependence precise, we introduce metrics that quantify superposition across experts and establish, consistently across configurations, that greater network sparsity yields greater *monosemanticity*. Building on this result, we advance a representational definition of expert specialization grounded in monosemantic feature representation rather than in load balancing, and we show that experts organize around coherent feature combinations whenever initialization permits it. Taken together, and within the boundaries of this controlled setting, these findings identify network sparsity—rather than feature sparsity—as the primary determinant of representational strategy in MoEs, and indicate that it confers more interpretable representations at only a small and diminishing cost in reconstruction loss. This evidence suggests that the assumed tension between interpretability and capability need not hold, at least for the class of toy models we study.

1 INTRODUCTION

Mixture-of-Experts (MoE) architectures now underpin state-of-the-art language models such as Qwen3, Mixtral, and Gemini (Yang et al., 2025a; Jiang et al., 2024; Google DeepMind, 2025), adopted for the favorable trade-off they strike between computational cost and performance (Shazeer et al., 2017; Fedus et al., 2022). A substantial body of subsequent work has refined the paradigm—sharpening routing through mechanisms such as Expert-Choice routing and strengthening training stability and transfer through approaches such as ST-MoE (Zhou et al., 2022; Zoph et al., 2022)—while theoretical and empirical analyses have shown that learnable routers can recover latent cluster structure in the data, furnishing a partial account of why experts come to specialize (Dikkala et al., 2023). Despite this widespread adoption and algorithmic maturity, however, MoEs remain comparatively opaque from a mechanistic-interpretability standpoint: we still lack a principled account of how their internal representations differ from those of the dense networks they increasingly displace.

Interpretability-oriented approaches have sought to render expert behavior more transparent. Yang et al. (2025b) propose MOE-X, which couples sparsity-aware routing with wide, ReLU-based experts in order to suppress polysemanticity, while Park et al. (2025) introduce MONET, scaling the number of experts so that individual capabilities can be edited through targeted expert activation. These efforts intervene primarily at the level of architecture,

and in doing so leave unresolved the more basic mechanistic questions on which their own motivation rests: how MoEs represent features, how the expert structure shapes superposition, and whether experts specialize of their own accord in the absence of additional regularization. That this gap is central rather than incidental is underscored by Mu & Lin (2025), whose survey of MoE research identifies mechanistic interpretability as a key open challenge. It is precisely these questions that we take up.

A fundamental obstacle to interpreting neural networks is the phenomenon of superposition: the representation of more features than there are dimensions. Superposition allows a network to pack many sparse features into a smaller set of neurons, but it does so at a direct cost to interpretability, rendering individual neurons polysemantic—each responding to a mixture of unrelated features rather than to a single, isolable one, and thereby severing the correspondence between neurons and the concepts one would hope to read off them.

MoE architectures introduce a further axis to this problem: *network sparsity*. Whereas a dense model activates every neuron regardless of input, an MoE activates only a fraction of its total parameters on any given input (Shazeer et al., 2017). This distinction is consequential for representation. A dense model must exploit feature sparsity by packing many sparse features into shared neurons; an MoE, by contrast, can afford to be selective, dedicating entire experts to specific feature combinations rather than compressing those combinations into a common subspace. Network sparsity thus reframes the representational problem from one of compression to one of allocation, and it is this reframing—together with its consequences for superposition, monosemanticity, and expert specialization—that our study renders precise within a controlled setting.

We organize our investigation around three questions: (1) whether MoEs exhibit less superposition than their dense counterparts; (2) whether the superposition of a given feature within an MoE expert undergoes a discrete phase change as a function of that feature’s relative importance and the overall feature sparsity, as it does in dense models; and (3) whether expert specialization is more faithfully understood through the lens of feature representation than through load balancing alone.

We address these questions within controlled toy models that extend the sparse-feature reconstruction framework of Elhage et al. (2022) to the MoE setting, and we advance three contributions, each demonstrated within that setting. **First**, we show that MoEs, unlike dense models, do not exhibit sharp phase changes; increasing network sparsity instead drives smooth, continuous transitions between representational regimes. **Second**, we establish that MoEs attain greater monosemanticity—equivalently, less superposition—than dense models matched on both active and total parameters, with individual experts representing features markedly more cleanly, and we quantify the accompanying reconstruction-loss cost as small and diminishing in the number of experts. **Third**, we introduce an interpretability-focused definition of expert specialization grounded in monosemantic feature representation, and demonstrate that experts organize around coherent feature combinations rather than around the arbitrary partitions that load balancing alone would produce. Collectively, and for the class of models we study, these results identify network sparsity—rather than feature sparsity—as the primary determinant of an MoE’s representational strategy.

2 RELATED WORK

Superposition and Feature Representations. The Linear Representation Hypothesis holds that networks encode concepts as directions in activation space (Park et al., 2024), yet the number of interpretable features routinely exceeds the available dimensions. Elhage et al. (2022) formalized the resolution of this tension as superposition, showing that dense models exploit non-orthogonal feature packing to maximize capacity at the cost of polysemanticity. Although superposition is a provably efficient use of capacity (Scherlis et al., 2025), it forces reliance on elaborate post-hoc disentanglement methods—most prominently Sparse Autoencoders—to recover monosemantic features (Bricken et al., 2023). Complementary work has characterized how data correlations shape superposition (Prieto et al., 2025) and how interference patterns arise and can be mitigated (Gurnee et al., 2023). We extend this line of inquiry to MoE architectures, and within the same toy-model framework

establish that it is network sparsity, rather than feature sparsity alone, that governs the representational strategy a model adopts.

Interpretability of MoEs. Although MoEs have become the standard for scaling large language models (Shazeer et al., 2017; Fedus et al., 2022; Jiang et al., 2024), mechanistic understanding of their internal representations continues to lag behind that of dense networks. Existing analysis concentrates on macroscopic behaviors—routing stability (Zoph et al., 2022), expert-choice statistics (Zhou et al., 2022), and latent cluster discovery (Dikkala et al., 2023)—rather than on the geometry of the representations themselves. More recent interpretability-oriented work seeks to impose specialization through architectural constraints, whether by predefining concept routing (Yang et al., 2025b) or by scaling expert counts to match vocabulary sizes (Park et al., 2025); these methods, however, prioritize capability editing over an account of intrinsic feature geometry. We address this gap directly, analyzing how the expert structure shapes superposition, establishing within a controlled setting that MoEs achieve greater monosemanticity than matched dense models, and proposing a feature-based definition of expert specialization that follows from this representational account.

3 BACKGROUND

A central aim of mechanistic interpretability is to reverse-engineer neural networks, and a productive strategy toward that aim is to decompose a model’s representations into a set of human-interpretable concepts termed ‘features’. Under the widely adopted linearity assumption, any hidden state h admits the decomposition

$$h = \sum_{i \in F} \alpha_i \vec{f}_i + \vec{b}$$

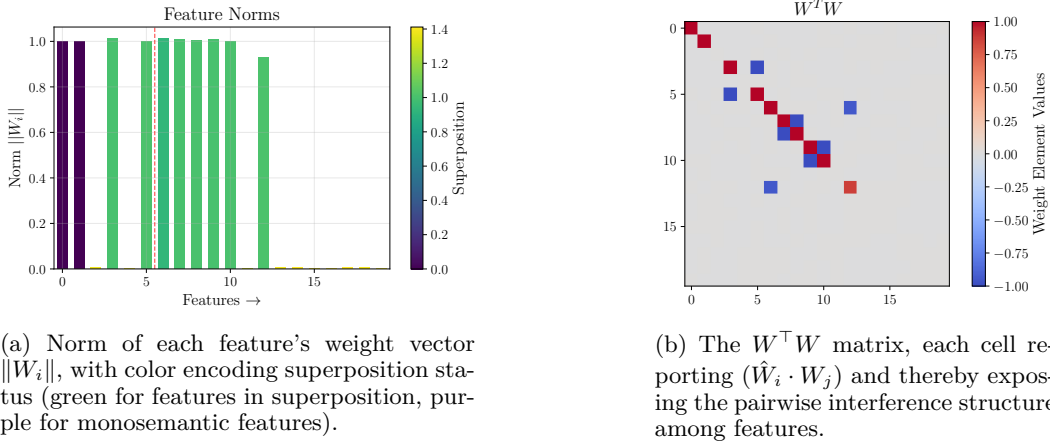
where $\vec{f}_i$ is the direction associated with feature i , the scalar α_i denotes that feature’s activation strength—informally, the degree to which feature i is present in the input—and F is the set of all represented features. This decomposition supplies the vocabulary in which superposition, interference, and monosemanticity are defined below.

The *superposition hypothesis* holds that a model can represent far more features than it has dimensions—that is, $|F| > m$ for $h \in \mathbb{R}^m$ (Elhage et al., 2022). Accommodating more features than dimensions within a latent space requires that the feature vectors in F be packed non-orthogonally, and whenever two features exhibit nonzero interference $\langle \vec{f}_i, \vec{f}_j \rangle \neq 0$ we say they are in *superposition*. Such packing is tolerable precisely when features are sparsely active—a property characteristic of most online text data—but it exacts a cost in interpretability, because the recovered activation α_i then carries spurious contributions unrelated to the presence of feature i in the input.

Monosemanticity, by contrast, is a property of individual neurons: a neuron is monosemantic when its activation corresponds cleanly to a single α_i —equivalently, when features are basis-aligned. The two notions are distinct. When features are orthogonal, and hence not in superposition, but are nonetheless not basis-aligned, neurons remain *polysemantic* even though feature interference is minimal. Our concern throughout is with reducing superposition rather than with enforcing basis-alignment; accordingly, when we describe a model, expert, or feature as “more monosemantic,” we mean precisely that it exhibits less superposition.

4 DEMONSTRATING SUPERPOSITION

MoEs are commonly conceptualized as compositions of dense models, with each expert taken to behave like an independent dense network. Whether experts in fact represent features as dense models do is, however, an open question, and one with direct bearing on how MoEs should be interpreted. We address it empirically, comparing the superposition exhibited by MoEs and by dense models under carefully matched conditions.



(a) Norm of each feature’s weight vector $\|W_i\|$, with color encoding superposition status (green for features in superposition, purple for monosemantic features).

(b) The $W^\top W$ matrix, each cell reporting $(\hat{W}_i \cdot W_j)$ and thereby exposing the pairwise interference structure among features.

Figure 1: Feature representation and superposition in a dense model with $n = 20$ features and $m = 6$ hidden dimensions, under importance $I = 0.7^i$ and uniform feature density $(1 - S) = 0.1$. The color-encoded superposition score is $\sum_j (\hat{W}_i \cdot W_j)^2$. Any feature may interfere with any other, so the interference structure is unrestricted—the reference against which the block-diagonal MoE structure of Figure 2 is to be read.

4.1 EXPERIMENTAL SETUP

Our objective is to characterize how an MoE projects a high-dimensional input $x \in \mathbb{R}^n$ into lower-dimensional expert representations $h \in \mathbb{R}^m$ and subsequently recovers it. The construction extends the sparse-feature reconstruction framework of Elhage et al. (2022) to the MoE setting, and it is precisely this synthetic construction—rather than any large-scale system—that affords controlled and exact access to superposition.

The input distribution. The input vector x stands in for the activations of an idealized, fully disentangled model in which each dimension x_i corresponds to a distinct, independent feature—a perfectly neuron-aligned, monosemantic representation. We draw $x_i \sim U(0, 1)$ subject to a sparsity level $S \in [0, 1]$, so that $P(x_i = 0) = S$; the features are thus independent sparse coordinates, an assumption we revisit as a scope restriction in Appendix A.6. Because features do not contribute equally to the objective, we assign each feature x_i a scalar importance I_i that reweights its term in the reconstruction loss. To isolate and study the phase transition of a single feature, we further scale the magnitude of the last feature by a factor $r \in \mathbb{R}^+$, so that $I = (1, 1, \dots, r)$. The importances I_i and the scale r jointly govern the signal strength of the features, allowing us to probe expert sensitivity to feature magnitude directly.

Model Architecture. The MoE comprises E experts, each expert e parameterized by a weight matrix $W^e \in \mathbb{R}^{m \times n}$ and a bias $b^e \in \mathbb{R}^n$. A learned router $g(x) = \text{softmax}(W^r x)$, with $W^r \in \mathbb{R}^{E \times n}$, assigns each input to its top- k experts. Every active expert first projects the input onto a lower-dimensional hidden state $h^e = W^e x$ and then produces a reconstruction $\hat{x}^e = \text{ReLU}((W^e)^\top h^e + b^e)$ ¹. The final output is the gated combination of the active experts, $x' = \sum_{e \in \text{top-}k} w_e \hat{x}^e$, where the w_e are the renormalized gating weights. Unless otherwise stated, we take $k = 1$, so that exactly one expert is active per input—the regime that most closely approximates a dense model and therefore furnishes the most conservative basis for comparison.

¹The output ReLU enforces non-negative reconstructions, matching our input distribution, in which $x_i \in [0, 1]$ whenever nonzero. It also interacts favorably with the interference structure: the off-diagonal terms of $(W^e)^\top W^e$ contribute negative interference, which the ReLU suppresses. When features are sparse, such negative interference becomes effectively “free,” since it is clipped to zero, thereby incentivizing configurations with negative off-diagonal terms, such as antipodal pairs.

We train all models to minimize an importance-weighted $L2$ reconstruction loss, $\mathcal{L} = \sum_x \sum_i I_i (x_i - x'_i)^2$. In the setting of Section 4, we augment this objective with the standard auxiliary load-balancing loss (Fedus et al., 2022), $\mathcal{L}_{\text{aux}} = \alpha N \sum_{e=1}^N f_e P_e$, in order to prevent expert collapse. Here N is the total number of experts, f_e is the fraction of a batch’s samples routed to expert e , P_e is the average gating probability assigned to expert e over the batch, and $\alpha = 0.01$ sets the penalty strength. In Section 4, by contrast, we deliberately omit this auxiliary term, so as to isolate the MoE’s intrinsic architectural bias toward or against superposition rather than confounding it with an explicit balancing objective; because load balancing is simultaneously a practical MoE ingredient and a potential confound for specialization, we treat it as an experimental variable and examine its effect directly.

4.2 MEASURING FEATURE CAPACITY

To analyze feature representations across architectures, we quantify two fundamental properties of every feature: its *representation strength* and its *interference* with other features. Representation strength within an expert e is measured by the norm $\|W_i^e\|$ of the feature’s weight vector, which registers the extent to which feature i is represented in that expert: $\|W_i^e\| \approx 1$ when the feature is fully represented, and $\|W_i^e\| \approx 0$ when it is not learned. Interference between feature i and the remaining features within expert e is read directly off the Gram matrix $W^\top W$, whose off-diagonal elements are exactly the pairwise interference terms.

As Figures 1a and 2a show, at equal total parameters ($m_{\text{dense}} = \sum m_{\text{experts}} = 6$) the dense model and the MoE represent comparable numbers of features—10 against 8—with similar norms. The decisive difference lies not in how many features are represented but in how interference is distributed among them. Although individual MoE experts exhibit some local superposition (Expert 0 in Figure 2b, for instance), the global interference structure is strictly partitioned: whereas in the dense model any feature may interfere with any other (Figure 1b), the MoE enforces a block-diagonal structure in which features routed to distinct experts do not interfere at all. In this controlled comparison, then, MoEs allocate representational capacity by partitioning the feature space itself, confining interference to within-expert blocks and eliminating it across them.

Expert Feature Dimensionality. To characterize how MoEs allocate their limited representational capacity differently from dense models, we turn to feature dimensionality—the “fraction of a dimension” that a given feature commands in a model (Elhage et al., 2022). For feature i , we define its dimensionality within expert e as

$$D_i^e = \frac{\|W_i^e\|^2}{\sum_j (\hat{W}_i^e \cdot W_j^e)^2} \quad (1)$$

The quantity D_i^e is bounded between zero (feature not learned) and one (fully monosemantic), so the total capacity of the MoE is accordingly $D = \sum_e \sum_{i=1}^n D_i^e$.

Efficient Packing. When features are “efficiently packed” within a model’s representation space, their dimensionalities sum to the number of embedding dimensions, $\sum_{i=1}^n D_i^e \approx m$ (Cohen et al., 2014; Scherlis et al., 2025; Elhage et al., 2022); for an MoE this generalizes to $\sum_e \sum_{i=1}^n D_i^e \approx E \cdot m$. Empirically, both dense and MoE models satisfy this constraint. The MoEs therefore pack features just as efficiently as dense models at equal total parameters, which establishes that their reduced superposition—documented below—is not purchased at the price of wasted capacity, but reflects a genuine reallocation of that capacity.

Features per Dimension & Network Sparsity. Because both dense and MoE models pack features efficiently, the number of features per dimension furnishes a direct and commensurate measure of superposition across architectures—and, in the MoE case, of how the expert count modulates superposition at each level of feature sparsity. A value exceeding one signals superposition, since the model then represents more features than it has dimensions. We define features per dimension for an MoE as

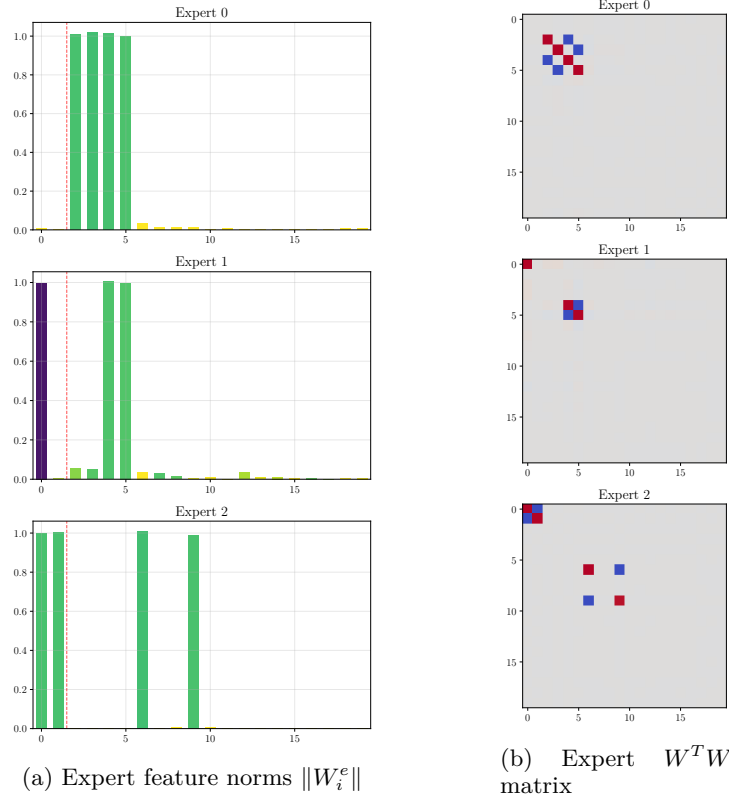


Figure 2: Feature representation and superposition in an MoE with $n = 20$ features, 3 total experts, and $m = 2$ hidden dimensions per expert under top- $k = 1$ routing, with importance $I = 0.7^i$ and feature density $1 - S = 0.1$. Relative to the dense model of Figure 1 at matched total parameters ($\sum_e m = 6 = m_{\text{dense}}$), the MoE represents a comparable number of features but confines all interference to within-expert blocks, so that cross-expert interference vanishes.

$$\frac{1}{k} \sum_{e=1}^E p_e \frac{\|W^e\|_F^2}{m} \quad (2)$$

where $\|W\|_F^2$ is the squared Frobenius norm and p_e is the expected usage probability of expert e across a batch—that is, its average renormalized gating weight after top- k routing.

At *equal* total parameters, the dense model consistently attains a higher features-per-dimension value than every MoE (Figure 3) and therefore carries more superposition: it exploits superposition to represent more features than it has dimensions ($n_{\text{learned}} > m_{\text{total}}$), whereas the MoE holds its representation at the monosemantic limit ($n_{\text{learned}} \approx m_{\text{total}}$). This effect is orderly rather than incidental. Increasing the number of experts, with total parameters and the ratio k/E held fixed, monotonically reduces features per dimension—the *more experts, the less superposition*—so that features become progressively more monosemantic as the expert count grows. The representational gain is bought at only a modest and shrinking cost in accuracy. The dense model’s greater superposition does permit it to achieve consistently lower reconstruction loss than the MoEs (Figure 6, Appendix A.1); the gap at any given sparsity is small, on the order of ~ 0.03 – 0.08 , and it narrows steadily as experts are added, so that MoEs with sufficiently many experts recover loss comparable to the dense baseline. The trade-off between reconstruction loss and monosemanticity is thus real but slight in this setting, and it moves in the MoE’s favor as network sparsity increases;

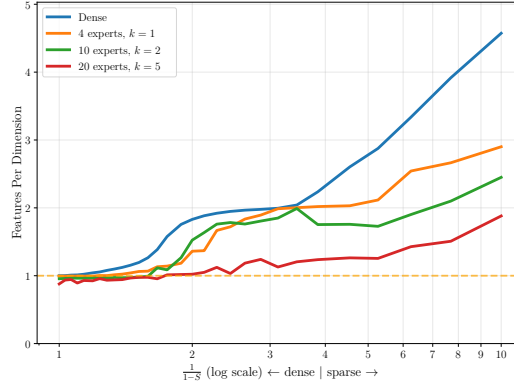


Figure 3: Features per dimension against inverse feature density ($\frac{1}{1-S}$) for **dense** and **MoE** architectures under uniform feature importance ($I_i = 1.0$). The dense model ($n = 100$, $m = 20$) exhibits the most superposition, which declines monotonically as the expert count grows: 4 experts with $m = 5$, $k = 1$ (orange); 10 experts with $m = 2$, $k = 2$ (green); and 20 experts with $m = 1$, $k = 5$ (red). Because all models are matched on total parameters and on the ratio k/E , this ordering isolates the effect of network sparsity alone: more experts yield fewer features per dimension, and thus less superposition. The dashed line at 1.0 marks the monosemantic limit, toward which the sparsest MoE converges.

Appendix A.2 supplies the intuition for why network sparsity produces this trade-off, and Appendix A.1 reports the loss curves in full.

5 PHASE CHANGE

Although MoEs and dense models learn a similar number of features, MoEs distribute those features across experts with substantially less interference. Network sparsity therefore reshapes how features are allocated rather than how many are learned. We now examine how the properties of the input distribution—feature sparsity and importance—drive this allocation, and whether they induce the physics-inspired *phase changes* that characterize dense models.

A model commands only a finite repertoire of ways to represent any feature: it may ignore the feature, place it in superposition, or represent it monosemantically. A phase change is the observation that the boundaries between these regimes can be discrete, surfacing as sharp transitions in the joint function of feature sparsity and relative importance.

Dense toy models are known to exhibit such discontinuous phase changes between internal feature representations (Elhage et al., 2022): varying the sparsity and relative importance of features in the input distribution elicits qualitatively different behavior, with greater feature sparsity, for instance, encouraging greater superposition. Analyzing the per-expert phase diagram of MoEs reveals that they adopt representational strategies distinct from those of dense models.

Setup. We adopt the setup of Section 4.1, augmenting it with the load-balancing loss to forestall specialization collapse—the failure mode in which experts and routers settle into local minima that leave certain experts entirely unused. We study three configurations, each with a single active expert ($k = 1$): (A) $n = 2$, $m = 1$; (B) $n = 3$, $m = 1$; and (C) $n = 3$, $m = 2$. For each, we report the expert-specific phase diagram over feature sparsity and last-feature relative importance while varying network sparsity, increasing the number of experts E up to the number of input feature dimensions n .

Here we fix active parameters rather than total parameters, so that $m_{dense} = km_{moe}$ (setting aside router parameters). This choice confines the comparison to the architecture itself: were active parameters permitted to differ, any observed effect could be attributed to their count

rather than to the MoE structure. As a consequence, 4.C.1/1 has the same number of active parameters as 4.B.X/2, yet the latter must encode the same $n = 3$ input features into a single hidden dimension ($m = 1$) rather than two ($m = 2$)—a constraint that leaves little room for superposition and correspondingly limits what this comparison can reveal about specialization.

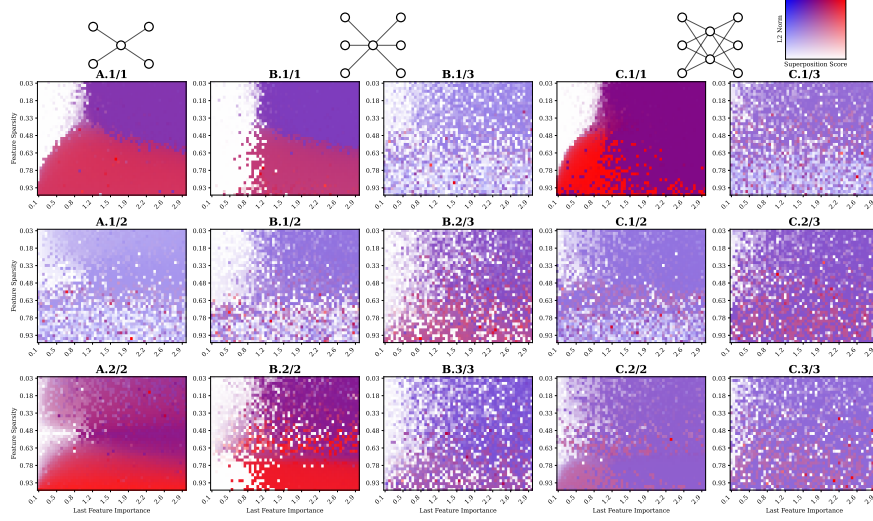


Figure 4: For a given expert and input feature, we decode how that feature is embedded in the hidden dimension—whether it is ignored (white), represented monosemantically (blue-purple), or placed in superposition (red). Each panel plots the joint of feature norm ($\|W_n\|^2$) and superposition score ($\sum_{j<n}(\hat{W}_n \cdot W_j)^2$) for the last of n features (subscript n) over feature sparsity $S \in [0.1, 1]$ and relative last-feature importance $I_n \in [0.1, 3]$; for each cell we train ten models and retain the lowest-loss run, applying the load-balancing loss throughout this section. A low $L2$ norm ($\|W_n\|$) renders the cell white, indicating that the model ignores the last feature; otherwise, a low superposition score renders it blue-purple, indicating monosemantic representation, and a high score renders it red, indicating superposition. Cell (i, j) of subfigure X.e/E reports expert e of E total experts trained on architecture X at last-feature importance $I_n = i$ and sparsity $S = j$; X.1/1 denotes the dense baseline. Read across increasing E , the sharp phase boundary of the dense model dissolves into continuous transitions.

Results and Takeaways. In every single-expert (dense) case we observe a clear phase change (Figure 4.X.1/1), reproducing the finding of Elhage et al. (2022). Increasing the number of experts, however, abolishes the discrete phase change altogether. Some experts in the $E = 2$ MoEs echo their corresponding dense cases (Figure 4.X.2/2) but transition continuously rather than sharply, and in each configuration one expert becomes markedly more monosemantic, specializing in the feature of greatest relative importance. Experts that do not resemble the dense case attain uniformly lower superposition scores (rendered bluer), signaling more monosemantic representations. This corroborates the preceding section: across the settings we examine, MoEs favor lower superposition than their dense counterparts.

In the $n = 2, m = 1$ setup (Figure 4.A), the dense model fails to represent the last feature at low feature sparsity, whereas the comparable MoE, possessing the capacity to do so, preserves it far more readily. With three input dimensions (Figure 4.B), the MoEs no longer enjoy this advantage, because the experts already place the other two features in superposition and leave no room for a third within a single hidden dimension; here alone the superposition-bearing hidden dimension is insufficient, in the high-sparsity regime, to represent all features. Even so, no clear phase change emerges, save for the 0.5–0.7 feature-sparsity region of 4.B.2/2, where the transition is largely discrete but mixed. For $m = 2$, the white region of the dense model (Figure 4.C.1/1), spanning the mid- to low-sparsity

domain in which the last feature is relatively less important than the others, indicates that this feature is ignored. As network sparsity increases across the remaining panels of Figure 4.C, the models represent the last feature with progressively greater $L2$ magnitude ($\|W_3^1\| < \|W_3^2\| < \|W_3^3\|$); equivalently, dimensionality in the low-importance region rises with network sparsity, in agreement with Figure 3.

Across Figures 4.B.2/2, 4.C.1/2, and 4.C.2/2 we observe a window of feature sparsity, roughly 0.48 to 0.7, within which polysemantic, monosemantic, and ignored representations are heavily intermixed. This intermediate regime reflects a genuine middle ground for MoEs, one in which polysemantic and monosemantic representations attain comparable loss and the model cannot commit consistently to the clean strategies it adopts in the low- and high-sparsity domains. Dense models exhibit no such regime—direct evidence that, in this setting, MoEs learn qualitatively different representational strategies.

Conclusion. These experiments all employ top- $k = 1$ routing, so that exactly one expert is active at any time—the case that most closely resembles a dense model. Even under this most conservative comparison, behavior diverges sharply from the dense baseline already at $E = 2$, including in regimes where the hidden dimension retains ample superposition capacity to represent every feature. We conclude that, within this toy setting, it is misleading to regard an MoE as a mere aggregation of dense models: by restricting each expert to a subset of the feature domain, the router fundamentally alters both the behavior and the learning dynamics of the experts.

6 EXPERT SPECIALIZATION

Having established that MoEs exhibit less superposition, we now turn to how these monosemantic features are organized within experts, and to what that organization implies for specialization.

Expert specialization in MoEs has traditionally been framed in terms of load balancing—the distribution of inputs across experts (Chaudhari et al., 2025). This framing, however, elides the natural intuition behind specialization: that an expert ought to be invoked precisely when the concepts in which it specializes are present in the input, and not otherwise.

We therefore advance a representational definition. We call an expert specialized when it *occupies* certain feature directions in the input space and represents those features relatively monosemantically. We show that these two conditions are directly correlated, and that their joint satisfaction gives rise to load balancing as a consequence rather than as an externally imposed objective—recasting a quantity usually enforced by an auxiliary loss as an emergent property of specialization.

Because we fix $k = 1$, the router partitions the feature space into convex cones, one per expert (see Appendix A.3). By the definition of a cone, $\forall s > 0, \vec{x} \in C \rightarrow s\vec{x} \in C$, where C is the set of points in the cone and s any positive scalar; hence if a feature vector $\vec{x}$ is routed to a given expert, every positive scaling $s\vec{x}$ is routed there as well. In this case we say that feature x is contained within expert e , and that expert e therefore *occupies* x .

We find empirically that small models distributing the input space across more experts tend to achieve lower loss (Appendix A.4). This prompts a sharper question: does the allocation of the input space to particular experts entail anything about the representations those experts come to learn? Our definition predicts that it does—that such allocation implies *monosemanticity*—and we confirm below that this correlation persists up to the largest toy models we study (e.g., $m = 10$).

For models with $n > 2$, we test this relationship in both directions. First, we ask whether initializing an expert to occupy certain features in the input space causes it to represent those features more monosemantically. Second, and conversely, we ask whether the features an expert comes to represent monosemantically are in fact the features it occupies in the input space.

When the gate matrix is initialized with ones along the main diagonal, each expert monosemantically represents the single feature it initially occupied, and that feature alone (Fig-

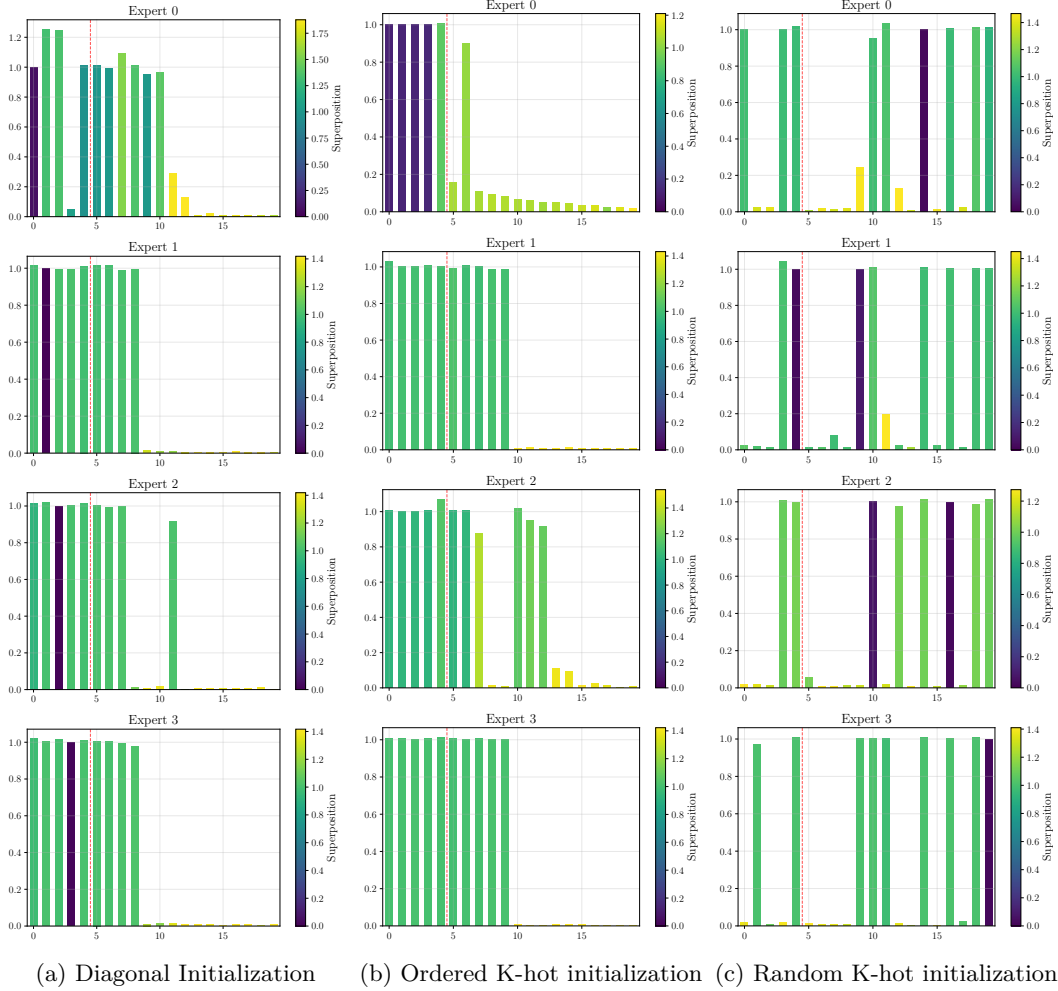


Figure 5: Expert feature norms $\|W_i^{(e)}\|$ and superposition (color) under three initialization schemes, with $n = 20, m = 5, E = 4, S = 0.1$. In (a), the gate matrix is initialized along the main diagonal ($W_i^T = \hat{e}_i$, the basis vector for that dimension), with relative feature importance decreasing exponentially in order from feature one to feature 20. In (b), the gate matrix is initialized “ordered k-hot”: the first expert aligns with the first five features and each subsequent expert with the next five, under the same importance ordering as in (a). In (c), the gate matrix is initialized “random k-hot”: each expert is assigned five features, drawn so that experts share none yet collectively cover all 20, with exponentially decreasing importance distributed randomly across features. Across all three schemes, the features an expert occupies at initialization are precisely those it comes to represent monosemantically.

ure 5a). Under ordered k-hot initialization, the first expert monosemantically represents four of the five features it initially occupied (Figure 5b); the remaining experts, initialized over lower-importance features, represent neither those less important features nor the five most important features over which they were not initialized. When we break the importance ordering and randomize the features each expert initially occupies, each expert monosemantically represents only the most important feature or features among those over which it was initialized (Figure 5c).

These results establish a strong correspondence between the features initially routed to an expert and the features it comes to represent monosemantically. Equally consistently, experts represent only important features monosemantically—whether we assign each expert

a single important feature explicitly, or supply it a set of features from which it then selects, unprompted, the most important one to represent cleanly.

Under uniform feature importance the effect softens but does not vanish: experts place all features in superposition with elevated interference, yet the features an expert initially occupies remain *relatively* more monosemantic, attaining superposition scores on average 1.03 standard deviations below that expert’s mean.

We next tested the converse direction: whether an expert that represents certain features monosemantically also occupies those features in the input. To this end we measure expert usage both when the features in question are *one of several* active features and when they are the *only* active feature—the latter being exactly the probability that the expert occupies those features. The correlation holds under both Xavier and k-hot initialization (Table 1). With $E = 10$, a mean expert usage near 10% reflects even load balancing across experts. In every case, activating an expert’s monosemantic feature or features raises its usage sharply, and restricting the input to those features alone makes the expert dominate usage outright. Most strikingly, under k-hot initialization 100% of the features an expert represents monosemantically are occupied by that same expert—an exact alignment between the two conditions of our specialization definition, and the clearest demonstration that representational specialization and input-space occupancy are two faces of a single phenomenon.

Table 1: Monosemantic-feature and usage statistics per expert for $n = 100, m = 10, E = 10$. For each initialization scheme (Xavier and k-hot) we train one hundred models, yielding 1000 experts per scheme; statistics are aggregated across models, with experts grouped by the number of features they represent monosemantically. For those features, we report expert usage when they are one of several active features in the input and, separately, when they are the *only* active feature—the latter measuring how strongly the expert occupies them. Conditional usage rises steeply as the number of monosemantic features grows, and approaches or reaches 100% once those features constitute the sole active input, confirming that monosemantic representation and input-space occupancy coincide.

Xavier Initialization				
Number of monosemantic features per expert	Number of experts (out of 1000)	Mean expert usage (%)	Mean expert usage; feature(s) active (%)	Mean expert usage; only feature(s) active (%)
0	461	—	—	—
1	387	9.595	17.94	67.18
2	138	9.599	30.29	95.65
3	13	8.363	40.19	100.0
4	1	1.428	14.69	100.0
5	0	—	—	—
K-Hot Initialization				
Number of monosemantic features per expert	Number of experts (out of 1000)	Mean expert usage (%)	Mean expert usage feature(s) active (%)	Mean expert usage only feature(s) active (%)
0	335	—	—	—
1	382	10.00	23.94	100.0
2	227	10.02	46.61	100.0
3	47	10.09	62.00	100.0
4	8	9.95	70.30	100.0
5	1	9.62	74.79	100.0

The more features an expert represents monosemantically, the more specialized it is: its usage on arbitrary input declines, while its usage conditional on its specialized features being active rises far above that of the other experts. This pattern holds in every case but one—the Xavier-initialized expert representing four monosemantic features, whose overall

utilization drops sharply—an isolated exception attributable to the fact that only a single such expert was observed across all 1000, and one that leaves the trend otherwise intact.

7 CONCLUSION

We set out to determine how the expert structure of an MoE shapes superposition, and within our controlled setting we arrive at three firm conclusions. MoEs consistently achieve greater monosemanticity than matched dense networks and, unlike dense models, exhibit no discrete phase change; it is network sparsity, rather than feature sparsity, that governs their representational strategy. On this basis we introduced a feature-based definition of expert specialization and showed that experts organize around coherent feature combinations whenever initialization permits it. These conclusions are drawn from controlled autoencoder toy models with synthetic, independently sparse features, and their transfer to large-scale transformers with unknown feature distributions remains to be established (Appendix A.6)—a boundary we regard as the natural next step rather than a peripheral caveat. Within these boundaries the evidence is nonetheless clear: toy MoEs attain reconstruction loss comparable to dense models while maintaining measurably more interpretable representations, which suggests that the assumed tension between mechanistic interpretability and model capability need not be fundamental, at least for the class of models studied here. Future work should identify what specifically favors monosemanticity in MoEs, how their training dynamics diverge from those of dense models, and when specialization emerges; it should also test whether these mechanisms survive under correlated or hierarchical features, richer routing, and objectives beyond reconstruction. Answering these questions would extend the present account toward the design of MoEs that are at once more interpretable and highly performant.

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A APPENDIX

A.1 MEASURING LOSS FOR VARYING SPARSITY & EXPERTS

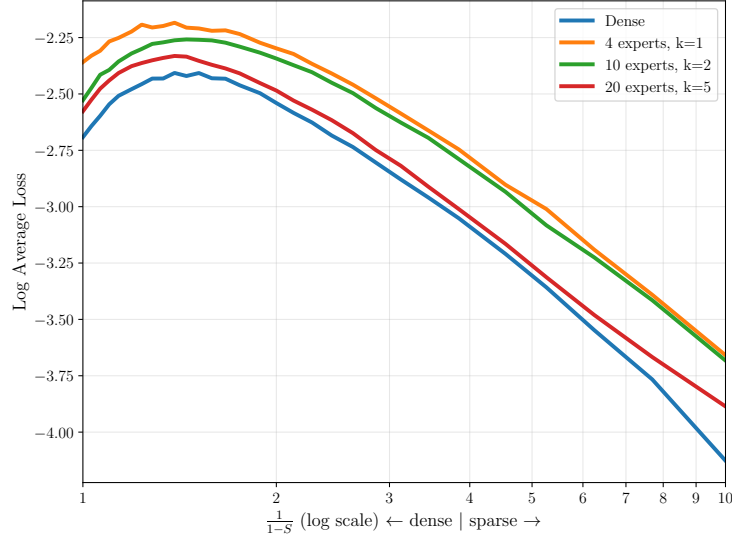


Figure 6: Log average loss against feature density ($\frac{1}{1-s}$) for dense ($m = 20$), MoE (4 experts, $k = 1$, $m = 5$), MoE (10 experts, $k = 2$, $m = 2$), and MoE (20 experts, $k = 5$, $m = 1$) models, all with uniform feature importance ($I_i = 1.0$) and $n = 100$ input features, averaged over five runs per sparsity level. The dense model attains the lowest loss at every sparsity level, yet the margin narrows steadily as the expert count grows: with sufficiently many experts the MoE loss converges toward the dense baseline. The interpretability gains reported in the main text therefore incur only a small and diminishing accuracy cost, and the loss-monosemanticity trade-off shifts in the MoE’s favor as network sparsity increases.

A.2 THEORETICAL INTUITION FOR SUPERPOSITION

In dense models, superposition arises in order to exploit the gap between sparse features and dense computation, compressing rarely active features into shared dimensions. MoEs close this gap structurally, through conditional computation: by aligning activation sparsity with feature sparsity, they remove the computational penalty of maintaining dedicated, rarely activating neurons. As Elhage et al. (2022) observe, once a model expends computation only on active features, splitting polysemantic neurons into dedicated monosemantic ones becomes the optimal strategy—in effect trading the *compression* afforded by superposition for the *selection* afforded by routing.

Geometrically, this structural shift manifests as a reduction in *interference*. The global ratio of represented features to parameters is unchanged, but the router partitions the feature space so that a feature routed to expert e competes for capacity only with the features likewise assigned to e . Interference is thus governed by the local Gram matrix $(W^e)^\top W^e$ rather than by the global $W^\top W$ of a dense model. By sharply reducing the number of features that can interfere with any given feature vector, this partitioning relaxes the optimization pressure to pack features in superposition and permits each expert to converge on monosemantic representations. This account is an intuition consistent with our measurements rather than a formal guarantee; the analytic construction of Appendix A.5 makes precise the one special case in which the correspondence can be exhibited exactly.

A.3 ROUTER SUBSPACES ARE CONVEX CONES

For $k = 1$, the router $g(x) = \text{softmax}(W^r x)$ reduces to $\text{argmax}(W^r x)$, so the region routed to expert i is exactly $\{x : \forall j \neq i, (w_i - w_j)^T x > 0\}$, where w_i and w_j are rows of W^r . Each constraint is a homogeneous linear inequality, and a region delimited by such inequalities is by definition a convex cone. Closure under positive scaling is immediate: if x satisfies the inequalities, so does sx for every $s > 0$. Closure under convex combination follows likewise: if x_1 and x_2 both satisfy them, then so does $x = \lambda x_1 + (1 - \lambda)x_2$ for every $\lambda \in [0, 1]$.

For $k > 1$, the set of inputs routed to a given expert e becomes a union of such convex cones, and a union of convex cones is in general not itself a convex cone. The account of experts as occupying feature directions therefore need not extend beyond $k = 1$ —a boundary we state explicitly rather than assume away, and one that circumscribes the specialization analysis of Section 6 accordingly.

A.4 EXPERT ROUTING WITH DIFFERENT INITIALIZATION SCHEMES

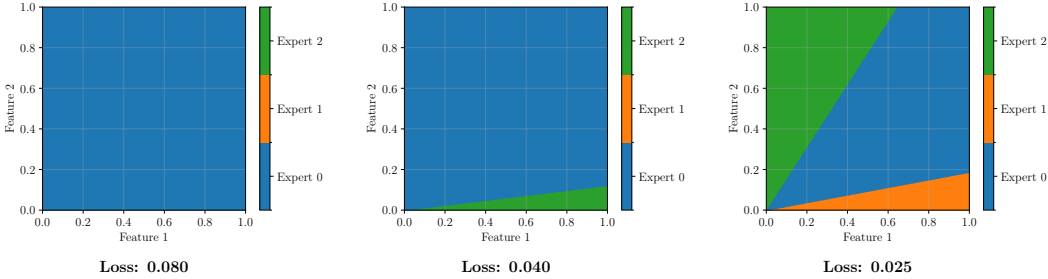


Figure 7: Expert routing of three architecturally identical models ($n = 2$, $E = 3$, $m = 1$) that differ only in initialization. The leftmost model performs worst (loss 0.08) and collapses all inputs onto a single expert. The middle model performs better (loss 0.04), routing a small portion of inputs—those for which feature 1 is active—to a second expert. The rightmost model attains the lowest loss (0.025) and distributes the input space across all three experts: one is selected when only feature 1 is active, one when only feature 2 is active, and one when both are active. Across these three models, lower loss coincides with a finer partition of the input space among experts.

In small models ($n = 2$, $m = 1$, $E > 1$) with $E \in [2, 7]$, we find empirically that models distributing the input space across more experts tend to achieve lower loss. Fixing $n = 2$ allows us to visualize directly which portions of the input space are routed to which experts, as in Figure 7.

A.5 ANALYTIC MODEL EQUIVARIANCE

For the toy setting of single-layer, single-nonlinearity, top- $k = 1$ MoEs, we can exhibit an explicit map from any dense model to a monosemantic MoE that represents an equivalent number of active features under a sparsity constraint. This construction is deliberately confined to that simplified setting; it establishes the correspondence exactly where it can be established, and does not purport to bound the behavior of the trained models studied in the main text.

Suppose the number of active features is bounded above by a for every input, i.e. $\forall x \in D : |\{i : x_i \neq 0\}| \leq a$, and suppose further that a does not exceed an expert’s hidden dimensionality m —an upper bound on how many features a single expert must represent—while m in turn does not exceed the total number of input features n , so that $a \leq m \leq n$. To construct the monosemantic MoE, assign to each subset $S \subseteq \{1, 2, \dots, n\}$ with $|S| \leq a$ an expert that preserves exactly those features monosemantically (it in fact suffices to take the subsets with $|S| = a$). The router then selects the expert corresponding to the active features, of which, by assumption, there are never more than a :

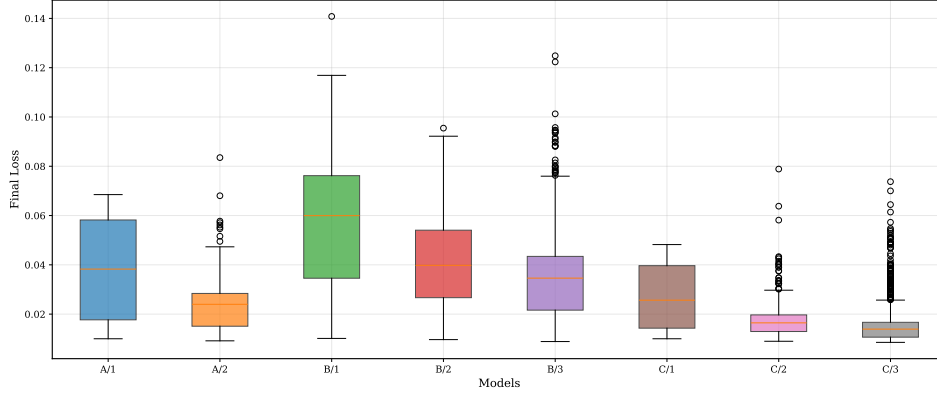


Figure 8: Model X/E denotes the architectures and models of Figure 4, with E the total number of experts (equivalently, the network sparsity). Increasing network sparsity lowers mean loss while raising localized variance, an effect most pronounced as the number of experts approaches the input feature dimensionality. We attribute the elevated variance to the comparatively unstable training of MoEs relative to dense models, which persists even though ten models are trained per cell and only the lowest-loss run is retained.

$$\text{Router}(x) = \arg \max_S \mathbb{I}[\text{support}(x) = S]$$

where $\text{support}(x) = \{i : x_i \neq 0\}$. Because $|S| \leq a \leq m$, every expert commands capacity sufficient to represent its assigned features without any superposition. In sum, at most a features are ever active, and each distinct combination of active features is handled by its own dedicated expert with the capacity to represent that combination monosemantically; the number of experts required is therefore $\binom{n}{m}$.

This construction reconstructs with exactly zero loss while the sparsity constraint holds (or, in these toy models, as sparsity tends to one), the qualification being that more than m features could in principle be active simultaneously ($a \not\leq m$), exceeding the network’s monosemantic capacity. Crucially, a dense polysemantic model fares no better in that event—absent feature correlations in the distribution—so even when $a \not\leq m$ occasionally, the monosemantic MoE confronts the same obstacle and may still outperform the dense model under looser sparsity constraints.

Consequently, for any dense model $f_{\text{dense}}(x) = \text{ReLU}(Wx + b)$ operating under the sparsity constraint $|\text{support}(x)| \leq a$, there exists an MoE $f_{\text{MoE}}(x)$ satisfying $f_{\text{dense}}(x) = f_{\text{MoE}}(x)$ for all valid inputs; in the toy settings of this paper the constraint holds in the limit of sparsity tending to one. In practice, moreover, the number of features that a fixed quantity of information can meaningfully encode is itself bounded—as reflected in the dimensionality of meaningful embeddings of that data—so that an MoE with sufficiently many experts and a tractable, still-interpretable degree of superposition may suffice to encode all features present.

A.6 LIMITATIONS

Our results should be read in light of several deliberate scope restrictions, which we state plainly so that the reach of our claims is not mistaken. First, all experiments are conducted in controlled toy-model settings adapted from superposition studies in sparse autoencoders (Elhage et al., 2022). This design is what makes precise measurement of interference and monosemanticity possible, but it abstracts away many of the complexities of large-scale transformers—multi-step computation, attention, heterogeneous and token-dependent feature distributions, and realistic routing dynamics—so that the transfer of our findings to production MoE architectures, and to trained sparse autoencoders extracted from them, remains to be established rather than assumed.

Second, our architectural and routing choices are intentionally constrained: simple linear/ReLU experts, fixed top- k routing (most often $k = 1$), matched parameter budgets between dense and MoE models, experts of identical capacity, and no auxiliary load-balancing loss except where noted. Practical MoEs employ richer routing, variable expert capacities, and multi-task objectives (Lepikhin et al., 2021; Fedus et al., 2022), or extend beyond reconstruction to objectives such as next-token prediction, any of which may induce qualitatively different representational behavior. Several of our conclusions may accordingly reflect these constraints in part, and disentangling the contribution of learned routing from that of a fixed block-sparse partition, or of expert width from expert count, is a direction we leave open.

Third, the feature distributions in our synthetic tasks—their sparsity patterns, importance profiles, and, in particular, the assumption of independent sparse coordinates—are markedly simpler than those of natural data, which are typically correlated, hierarchical, or distributed; whether the observed mechanisms persist under such structure is untested here.

Finally, although we observe increased monosemanticity and reduced superposition under our fixed conditions, and quantify the accompanying reconstruction-loss gap as small and diminishing in the expert count (Appendix A.1), we do not evaluate downstream task performance, training stability, or specialization dynamics at scale, which prior work indicates can shift substantially with model size, optimization regime, and data diversity (Krajewski et al., 2024). The interpretability gains we report are therefore established as representational-geometry gains under matched capacity, not as demonstrated improvements in downstream behavior.

In sum, our study yields clear mechanistic insight under clean experimental conditions, and validating these patterns in large, realistic MoE systems is the natural next step. Because the present methods do not scale directly to such models, developing scalable analogues—and applying the proposed specialization metric to activations from trained MoE language models—is itself a well-defined direction for future work.